

MECHANICAL ENGINEERING MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20
- Which of the following material has maximum ductility- i) Mild steel ii) Copper iii) Nickel iv) Aluminium
 - ☒ An eutectoid steel consists of- i) wholly pearlite ii) wholly austenite ☒ iii) pearlite and ferrite iv) pearlite and cementite
 - ☒ The hardness of steel depends upon the- i) amount of cementite it contains ☒ ii) amount of carbon it contains iii) contents of alloying elements iv) method of manufacture of steel
 - Blast furnace is used to produce- i) ☒ pig iron ii) steel iii) wrought iron iv) cast iron
 - The strength is the ability of a material to resist- i) ☒ deformation under stress ii) externally applied forces with breakdown or yielding iii) fracture due to high impact loads iv) none of these
 - ☒ In induction hardening _____ is high. i) ☒ current ii) voltage iii) frequency iv) temperature
 - Which of the following is an amorphous material- i) Mica ii) Glass iii) Lead iv) Silver
 - ☒ The type of space lattice found in alpha-iron is- i) FCC ☒ ii) BCC iii) HCP iv) None of these
 - i) The toughness of a material _____ when it is heated. i) remains same ☒ ii) increase iii) decrease
 - j) Muntz metal (Yellow brass) contains i) 70% copper and 30% zinc ii) 60% copper ☒ and 40% zinc iii) 59% copper, 40% zinc and 1% tin iv) 60.45% copper, 35.2% zinc and 5.35% nickel
 - ☒ Vanadium when added to steel- i) decreases tensile strength ii) raises critical temperature iii) Sowers critical temperature ☒ iv) increases tensile strength
 - ☒ i) increases tensile strength i) increases tensile strength ii) increases tensile strength iii) increases tensile strength ☒ iv) increases tensile strength
 - m) Which of the following property is desirable in parts subjected to shock and impact loads i) ☒ Strength ii) Brittleness iii) Stiffness iv) Toughness
 - n) The value of Atomic packing factor for BCC crystal structure is- i) 0.68 ii) 0.74 ☒ iii) 0.29 iv) none of these
 - ☒ o) Monel metal contains- i) 65% nickel, 15% chromium and 20% iron ii) ☒ 68% nickel, 29% copper and 3% other constituents iii) 80% nickel and 20% chromium iv) 80% nickel, 14% chromium and 6% iron

GROUP-B

2. ☒ a) Write short notes on - a) Bainite b) Spheroidite c) Martensite 6
☒ b) Write down few differences between Case Hardening & Flame Hardening 2
3. ☒ a) Write down composition of a) Munz metal b) Gun Metal c) Duralumin d) Y-Alloy 4
☒ b) What is Corrosion? Name few corrosion resistive materials? Mention two corrosion preventive methods. 4
4. ☒ a) Determine the average number of atoms, relation between lattice parameter and atomic radius, atomic packing factor of FCC crystal structure. 2+2+2
☒ b) Describe the process of Nitriding. 2

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| 5. | a) | Give one application for each of the followings - a)Acrylics b)Polyesters c)Epoxies d)Melamine | 4 |
| | b) | Briefly explain Dye penetration test. | 4 |
| 6. | a) | Describe Annealing process. State it's Purpose. | 4+2 |
| | b) | What is Martempering? | 2 |
| 7. | a) | Briefly explain a) Eutectic reaction b) Eutectoid reaction c) Peritectic reaction | 2+2+2 |
| | b) | Write down differences between White cast iron and Grey cast iron. | 2 |
| 8. | a) | Write down composition of Duralumin and Magnalium. | 2 |
| | b) | Write down one application each for laminated materials and fibre reinforced materials. | 2 |
| | c) | What do u mean by Vulcanization of Rubber? State its application. | 2+1 |
| 9. | a) | Write down differences between Hardness and Toughness. | 2 |
| | b) | What is powder metallurgy process? Write down few application of this process | 2+1 |
| | c) | Classify plain carbon steel and also mention their respective carbon percentage. // | 2+1 |

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